VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB with Java

Branch/ Class: B.Tech/M.Tech **Date:**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: **Reg. no.:**

Project Assessment

Important:

1. **All table values should be changed. No same value should present in other lab sheet.**
2. **All package name, class name, method name, variable name, field name in mongo db documents should be maintained as it is.**
3. **Need screenshot of all codes and outputs. No copy paste of code.**
4. **All screenshot should have date and time at the task bar.**
5. **No comment lines should be included in the code. Else it will be considered as AI Code**
6. **Don use any AI tools to take the code.**

GENERAL INSTRUCTIONS: Please carefully read the below instructions

The objective of this assessment is to check your ability to complete a project as per the provided "Project Design".

You are expected to –

1. Write the source code for the classes, methods and packages **EXACTLY** as mentioned in the "**Project Design**" section.
2. Ensure that the names of the packages, classes, methods and variables **EXACTLY MATCH** with the names specified in the "Project Design" section.
3. Understand the project requirements and ACCORDINGLY WRITE the code and logic in the classes and methods so as to meet all given requirements.

Creating the project and testing it –

1. You are expected to create your project locally using eclipse (or any other IDE) on your desktop.
2. Submit the screenshot of all the codes.
3. Should take the output screenshot for each test case given at the end.

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House Rental Management System

Project Objective:

Create a console-based Java application that would allow a user of a House Management System to perform the following operations

- Create a new Rental Property into the system [Posting availability of a new property that is available for rent]
- Find the details of all the rental properties that would satisfy the search criteria

Project Design:

A. Database Design:

1. **Create Collection [To be done using mongodb commands]**

Collection Name: RENTAL_TBL (All table values should be changed. No same value should present in other lab sheet)

Insert some sample records into the table:

PROPERTYID	RENTALAMOUNT	NOOFBEDROOMS	LOCATION	CITY
CHE1101	20000	2	AnnaNagar	Chennai
BEN1102	20000	2	HSR	Bengaluru
CHE1103	8000	1	Tambaram	Chennai
BEN1104	25000	2	HSR	Bengaluru
CHE1105	12000	2	Annanagar	Chennai

B. System Design:

Name of the package	Usage
com.vit.hms.service	This package will contain the class that displays the console menu and takes the user input.
com.vit.hms.bean	This package will contain the bean class
com.vit.hms.dao	This package will contain the class that will do the database related operations
com.vit.hms.util	This package will contain a class to establish database connection and also another class that handles the user defined exception.

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Package: com.vit.hms.util

Class	Method and Variables	Description
DBUtil		DB connection class
	public static Connection**getDBConnection**()	Establish a connection to the database and return the java.sql.Connection reference
InvalidCityException		User defined Exception class
	public String toString()	Returns "**INVALID CITY**"

Package: com.vit.hms.bean

Class	Method and Variables	Description
RentalPropertyBean		**Class**
	private float rentalAmount;	
	private int noOfBedRooms;	
	private String location;	
	private String city;	
	private String propertyId;	
	setters & getters	Should create the getter and setter methods for all the attributes mentioned in the class

Package: com.vit.hms.dao

Class	Method and Variables	Description
RentalPropertyDAO		DAO class
	public String generatePropertyID(String city)	This method should return the propertyId which is auto generated using Math.Random method. **Format:** First 3 letters of the city in uppercase followed by the 4 digit auto generated number.

		E.g. CHE1000 – if the city name is Chennai
	public int createRentalProperty(RentalPropertyBean bean)	This method inserts the bean's properties into the rental_tbl and returns the count of the records inserted Note: The propertyId should be auto generated first by invoking the appropriate DAO method
	public List<RentalPropertyBean> findPropertyByCriteria(float minRentalAmount,float maxRentalAmount,RentalPropertyBean bean)	This method should return the list of all the properties that satisfies the criteria given. Note: The location and city are case insensitive For e.g. the user may give location and city in any case (upper/lower) and it should return the list of properties that matches the cases irrespective of their case Example: minRentalAmount=12000 maxRentalAmount=18000 noOfBedRooms=2 location=Adyar city=Chennai This function should return a list that contains all the houses whose rental amount lies between 12000 and 18000 with 2 bedrooms that are available in Adyar, Chennai

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Package: com.vit.hms.service

Class	Method and Variables	Description
RentalPropertyService		Main class
	public static void **main**(String[] args)	The code that is needed to test your program goes here. A sample code is shown at the end of the document.
	public String addRentalProperty(RentalPropertyBean bean)	• It returns "**NULL VALUES IN INPUT**" if city or location is null • It returns "**INVALID INPUT**" under the following conditions ➢ Length of city or location is 0 ➢ No of bedrooms/Rental amount is 0 • This method will call validateCity method for validating city. If city is not either Chennai or Bengaluru (not case sensitive) the method should return "**INVALID CITY**" In all other conditions this method should call createRentalProperty method of DAO and returns "**SUCCESS with Your Regno**" if record is inserted else returns "**FAILURE**" if the record cannot be inserted
	public List<RentalPropertyBean> getPropertyByCriteria(float minRentalAmount,float maxRentalAmount,int noOfBedRooms,String location,String city)	• This method takes all the values received, creates a RentalPropertyBean Object and initializes it with it • It then invokes the findPropertyByCriteria method of the DAO class and returns the list received. • The returned records should match all the criteria provided by the user.
	public String	• It returns "**INVALID VALUES**" under

	fetchRentalProperty(float minRentalAmount,float maxRentalAmount,int noOfBedRooms,String location,String city)	the following conditions ➢ minRentalAmount/maxRental Amount is 0 ➢ maxRentalAmount is lesser than minRentalAmount ➢ noOfBedrooms is less than or equal to 0 ➢ location/city is null ➢ length of location/city is 0 • This method will call validateCity method for validating city. If city is not either Chennai or Bengaluru (not case sensitive) the method should return "**INVALID CITY**" • If none of the above conditions are matching, it invokes getPropertyByCriteria method of this class and receives the list • If there are no records in the list received, this method will return "**NO MATCHING RECORDS**" • Else this method would return a string in the following format **RECORDS AVAILABLE:5** [Here, 5 is the size of the list received]
	public void validateCity(String city) throws InvalidCityException	• This method will throw InvalidCityException if the provided city name is not either chennai or bengaluru (case insensitive)

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Test Cases:

Below are the actual set of test cases that the CPC test engine will run in the background. Please ensure that the conditions mentioned in these test-cases are handled by your class design.

1. Test to check if the newly generated property id is valid
2. Test to check if a new property with valid values are getting added successfully
3. Test to check if the DAO method returns the correct list of data (list with records)
4. Test to check if the DAO method returns the correct list of data (list with no records)
5. Test to check if the fetchRentalProperty method of service class returns expected values for different conditions

Main Method:

You can write code in the main method and test all the above test cases. A sample code of the main method to test a test case is shown below for your reference.

```
public static void main(String[] args) {

String result=new
RentalPropertyService().fetchRentalProperty(3000,15000,2,"Adyar","Chennai");

System.out.println(result);

}
```